EASM-Based Probabilistic Optimization of Large DOF Structural Systems under Seismic Loading

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The Enhanced Analytical Spectral Moments (EASM) method, developed by the authors in previous studies [1], provides an efficient probabilistic framework for seismic response evaluation based on random vibration theory. Ground motion is modeled through power spectral density functions consistent with design response spectra, while closed-form approximations of spectral moments enable a rapid estimation of response statistics without resorting to time-history analyses [2, 3]
Building on this framework, the EASM method is extended to the context of probabilistic structural optimization of large multi-degree-of-freedom systems. Owing to its computational efficiency, the approach is particularly well suited for iterative optimization procedures, where repeated evaluations of structural response are required. Response measures, such as expected peak displacements, accelerations, and internal forces, are directly incorporated into objective functions and constraints, allowing seismic uncertainty to be explicitly accounted for within the optimization process. The proposed framework also supports the assessment of alternative design strategies, including modifications of structural properties and the potential integration of vibration mitigation solutions, within a unified probabilistic setting.
The methodology is demonstrated through the application to a real building characterized by a high number of degrees of freedom, representative of large and complex structural systems. The results highlight the capability of the EASM-based approach to provide accurate response estimates with limited computational effort, enabling efficient performance-based optimization under seismic loading. The study confirms the potential of the EASM method as a powerful tool for the probabilistic analysis and optimization of large-scale structures, with direct implications for advanced seismic design and assessment.

References

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